

Experiment 6:

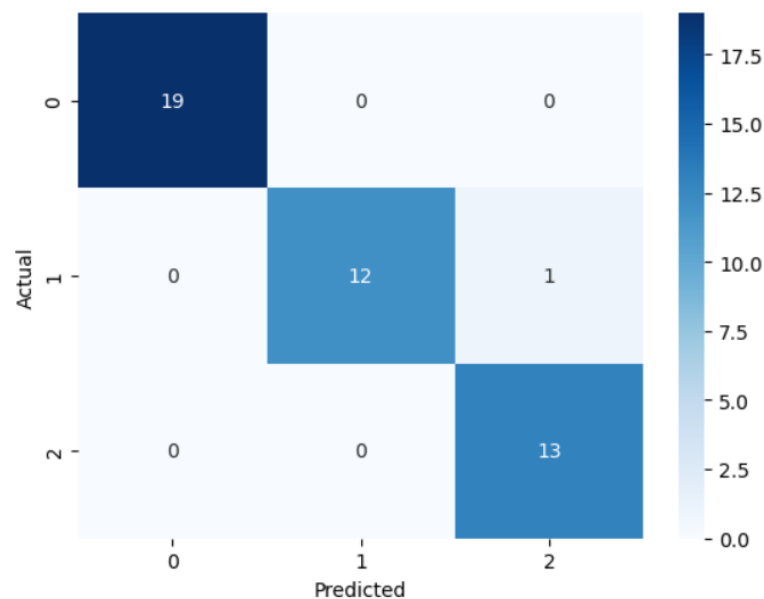
Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

Python Code:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score
import seaborn as sns
import matplotlib.pyplot as plt

X, y = load_iris(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
y_pred = GaussianNB().fit(X_train, y_train).predict(X_test)
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Output:



Accuracy: 0.9777777777777777